

Backend Development

Backend development refers to the server-side part of an application that handles business logic, database operations, authentication, APIs, and communication between the frontend and data sources.

Responsibilities

Processing requests, managing databases, business logic, APIs, security, and server configuration.

Backend Architecture

Frontend -> API Layer -> Business Logic Layer -> Data Access Layer -> Database

Backend Components

API Layer, Business Logic Layer, Data Access Layer, and Database Layer.

Common Technologies

Java, Python, Node.js, C#, PHP, Go with frameworks such as Spring Boot, Django, Express.js, ASP.NET Core, and Laravel.

API Development

REST APIs using GET, POST, PUT, and DELETE methods.

Database Integration

Controller -> Service -> Repository -> Database flow.

Authentication and Authorization

JWT, OAuth 2.0, MFA, and role-based access control.

Security Best Practices

Input validation, HTTPS, password hashing, rate limiting, XSS/CSRF protection.

Error Handling

Proper logging and user-friendly error responses.

Development Lifecycle

Requirements -> Design -> Implementation -> Testing -> Deployment -> Maintenance.

Testing

Unit, Integration, API, and Performance testing.

Deployment Architecture

Users -> Load Balancer -> App Server -> Database -> Cache Server.

Benefits

- Centralized data management
- Enhanced security
- Scalability
- Faster processing
- Reliable business logic
- External service integration